A Crash Course In Arduino Programming
Lesson 3 Exercises

1. You have an array of length 3, and you try to access the element at position 5. Do you think your code will compile? Write a program to attempt this and explain what happens.

2. Make an array of any length, but don’t specify the values (see the first part of method 1 in your handout). Try to access and print an element of the array. What happens?

3. The Arduino Uno has limited memory. Arrays make it easier for us to store data, but we can overshoot the memory capabilities of our microcontrollers if we’re not careful. How big of an array do you think an Arduino Uno can handle? How can you avoid using too much memory?

4. Errors with I2C communication can be hard to debug because your code will still compile (no error messages!). Take a piece of working I2C code and create problems one at time (maybe you can remove a Wire.write(), or change a correct register to an incorrect register). Explain the problems you created, the behavior of your program/circuit after you introduced the problems, and how you might recognize these problems if you create them by mistake in the future.